

AI 3000: Reinforcement Learning

Tutorial 1 Solutions

1 Markov Chain

1.1 (List the state space S of the Markov chain.) The state space consists of all possible states of the student:

$$S = \{\text{Hostel}, \text{Class 1}, \text{Class 2}, \text{Class 3}, \text{Pass}, \text{Sleep}, \text{Sports}\}.$$

We can abbreviate the states as $H, C_1, C_2, C_3, P, Sl, Sp$.

1.2 (Identify the absorbing state(s), if any. Justify your answer.) A state s is called **absorbing** if $P(s, s) = 1$. Equivalently, once the process enters s , it can never leave s .

From the graph, once the student reaches *Sleep*, the process remains in *Sleep*. Therefore, $P(\text{Sleep}, \text{Sleep}) = 1$. Hence, *Sleep* is an absorbing state. The other states have at least one transition to a different state.

1.3 (Suppose the student is currently in Class 2. List all possible next states together with their one-step transition probabilities.) From the transition graph, from *Class 2* we have

$$P(C_3 | C_2) = 0.8, \quad \text{and} \quad P(Sl | C_2) = 0.2.$$

Therefore, the possible next states are

C_3 with probability 0.8, Sl with probability 0.2.
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Thus, $P(C_3 | C_2) + P(Sl | C_2) = 0.8 + 0.2 = 1$.

1.4 (Is it possible for the student to move from *Hostel* to *Class 2* in one step? Explain.) From the graph, the only transitions out of *Hostel* are

$$P(H | H) = 0.9, \quad P(C_1 | H) = 0.1.$$

There is no direct transition

$$H \longrightarrow C_2.$$

Therefore,

$$P(C_2 | H) = 0.$$

The student can reach *Class 2*, but it requires at least two steps: $H \longrightarrow C_1 \longrightarrow C_2$. For example,

$$P(X_2 = C_2 | X_0 = H) = P(C_1 | H)P(C_2 | C_1) = 0.1(0.5) = 0.05.$$

1.5 (Starting from the *Sports* state, list all possible one-step successor states.)

From the *Sports* state, the possible transitions are

$$P(C_1 | Sp) = 0.2, \quad P(C_2 | Sp) = 0.4, \quad \text{and} \quad P(C_3 | Sp) = 0.4.$$

Therefore, $Sp \rightarrow C_1, C_2, C_3$ with probabilities 0.2, 0.4, 0.4. Check: $0.2 + 0.4 + 0.4 = 1$.

1.6 (Construct the transition probability matrix of the Markov chain.) Choose the state ordering ($H, C_1, C_2, C_3, P, Sl, Sp$). The transition probability matrix is

$$P = \begin{pmatrix} 0.9 & 0.1 & 0 & 0 & 0 & 0 & 0 \\ 0.5 & 0 & 0.5 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.8 & 0 & 0.2 & 0 \\ 0 & 0 & 0 & 0 & 0.6 & 0 & 0.4 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0.2 & 0.4 & 0.4 & 0 & 0 & 0 \end{pmatrix}.$$

Here,

$$\mathbb{P}_{ij} = P(X_{n+1} = j | X_n = i).$$

Important:

Always state the ordering of the states before writing a transition matrix. Otherwise the matrix is ambiguous.

1.7 (Verify that your transition probability matrix is stochastic.) A transition probability matrix must satisfy: $P_{ij} \geq 0$ for every i, j , and

$$\sum_j P_{ij} = 1$$

for every row i .

For example, the row corresponding to C_2 is $(0, 0, 0, 0.8, 0, 0, 0.2)$, and therefore $0.8 + 0.2 = 1$. Similarly, for the *Sports* state, $0.2 + 0.4 + 0.4 = 1$.

All rows satisfy the condition. Hence, P is a stochastic matrix.

1.8 (Starting from *Hostel*, give one possible path that reaches the *Sleep* state.)

One possible path is $H \rightarrow C_1 \rightarrow C_2 \rightarrow Sl$. The probability of this particular path is

$$\begin{aligned} \mathbb{P}(X_0 = H, X_1 = C_1, X_2 = C_2, X_3 = Sl) \\ &= P(C_1 | H)P(C_2 | C_1)P(Sl | C_2) \\ &= (0.1)(0.5)(0.2) \\ &= \boxed{0.01}. \end{aligned}$$

Thus, this is one possible trajectory that reaches Sleep.

1.9 (Explain why this process satisfies the Markov property) The Markov property says that the conditional distribution of the next state depends only on the current state:

$$\mathbb{P}(X_{n+1} = x_{n+1} \mid X_n = x_n, \dots, X_0 = x_0) = \mathbb{P}(X_{n+1} = x_{n+1} \mid X_n = x_n).$$

For example, consider the trajectory

$$H \rightarrow C_1 \rightarrow C_2.$$

Suppose the current state is C_2 . Regardless of how we arrived at C_2 ,

$$P(X_{n+1} = C_3 \mid X_n = C_2, \text{past}) = 0.8.$$

Similarly, $P(X_{n+1} = Sl \mid X_n = C_2, \text{past}) = 0.2$.

Therefore, the past trajectory does not provide additional information about the next state once the current state is known.

Hence,

$$\{X_n\}_{n \geq 0} \text{ satisfies the Markov property.}$$

1.10 (Simulation) The initial distribution is $\mu = (1, 0, 0, 0, 0, 0, 0)$. Therefore, $X_0 = H$ with probability 1.

To generate X_{n+1} , sample

$$U_n \sim \text{Unif}(0, 1)$$

and use the cumulative probabilities corresponding to the current state.

For example, suppose

$$U_0 = 0.05, \quad U_1 = 0.95, \quad U_2 = 0.70, \quad U_3 = 0.80, \quad U_4 = 0.50.$$

Then:

Current state	U_n	Next state
H	0.05	H
H	0.95	C_1
C_1	0.70	C_2
C_2	0.80	C_3
C_3	0.75	P

Thus one possible sample path is

$$H \rightarrow H \rightarrow C_1 \rightarrow C_2 \rightarrow C_3 \rightarrow P.$$

The simulation procedure is:

1. Sample $X_0 \sim \mu$.
2. Given $X_n = i$, look at row i of P .
3. Compute the cumulative probabilities.
4. Generate $U_n \sim \text{Unif}(0, 1)$.
5. Select $X_{n+1} = j$ according to the interval containing U_n .
6. Repeat until X_5 is generated.

For example, from Hostel,

$$P(H \mid H) = 0.9, \quad P(C_1 \mid H) = 0.1.$$

Hence,

$$U_n \in [0, 0.9) \Rightarrow X_{n+1} = H,$$

and

$$U_n \in [0.9, 1) \Rightarrow X_{n+1} = C_1.$$

Remember:

Simulation means repeatedly sampling from the *current* row of the transition matrix.

2 Markov Decision Processes

We use the state ordering $\mathcal{S} = \{A, B\}$ and action set $\mathcal{A} = \{L, R\}$.

From the given MDP:

	A	B
(A, L)	0.8	0.2
(B, L)	0.6	0.4
(A, R)	0.3	0.7
(B, R)	0.1	0.9

That is,

$$\begin{aligned}
 P(A \mid A, L) &= 0.8, & P(B \mid A, L) &= 0.2, \\
 P(A \mid B, L) &= 0.6, & P(B \mid B, L) &= 0.4, \\
 P(A \mid A, R) &= 0.3, & P(B \mid A, R) &= 0.7, \\
 P(A \mid B, R) &= 0.1, & P(B \mid B, R) &= 0.9.
 \end{aligned}$$

2.1 (Stationary Randomized Markov policy) The policy is

$$\pi(L \mid A) = 0.7, \quad \pi(R \mid A) = 0.3.$$

1. We want $P^\pi(S_{n+1} = B \mid S_n = A)$. Condition on the action selected at state A :

$$\begin{aligned} P^\pi(B \mid A) &= \sum_{a \in \{L, R\}} \pi(a \mid A) P(B \mid A, a) \\ &= \pi(L \mid A) P(B \mid A, L) + \pi(R \mid A) P(B \mid A, R) \\ &= (0.7)(0.2) + (0.3)(0.7) \\ &= 0.14 + 0.21 \\ &= \boxed{0.35}. \end{aligned}$$

Therefore,

$$\boxed{P^\pi(S_{n+1} = B \mid S_n = A) = 0.35.}$$

2. We want $P^\pi(S_{n+1} = A \mid S_n = B)$. At state B ,

$$\pi(L \mid B) = 0.4, \quad \pi(R \mid B) = 0.6.$$

Therefore,

$$\begin{aligned} P^\pi(A \mid B) &= \sum_{a \in \{L, R\}} \pi(a \mid B) P(A \mid B, a) \\ &= (0.4)(0.6) + (0.6)(0.1) \\ &= 0.24 + 0.06 \\ &= \boxed{0.30}. \end{aligned}$$

Hence,

$$\boxed{P^\pi(S_{n+1} = A \mid S_n = B) = 0.30.}$$

3. Induced State Transition Matrix: We have obtained $P^\pi(B \mid A) = 0.35$, so $P^\pi(A \mid A) = 1 - 0.35 = 0.65$.

Similarly, $P^\pi(A \mid B) = 0.30$, so $P^\pi(B \mid B) = 1 - 0.30 = 0.70$. Therefore, with state ordering (A, B) ,

$$\boxed{P^\pi = \begin{pmatrix} 0.65 & 0.35 \\ 0.30 & 0.70 \end{pmatrix}.}$$

Key formula:

For a Markov policy,

$$P^\pi(s'|s) = \sum_{a \in \mathcal{A}} \pi(a|s)P(s'|s, a).$$

The policy “averages” the environment transition probabilities according to the action probabilities.

4. We want $P^\pi((B, R) \mid (A, L))$. This event consists of two steps:

$$(A, L) \longrightarrow B$$

followed by selecting action R at state B .

Therefore,

$$\begin{aligned} P^\pi((B, R) \mid (A, L)) &= P(B \mid A, L)\pi(R \mid B) \\ &= (0.2)(0.6) \\ &= \boxed{0.12}. \end{aligned}$$

Important:

For the state-action process,

$$P((s', a') \mid (s, a)) = P(s' \mid s, a)\pi(a' \mid s').$$

First the environment produces s' , then the policy chooses a' .

Define the state-action process $Y_n = (S_n, A_n)$. The state space of Y_n is

$$\mathcal{S} \times \mathcal{A} = \{(A, L), (A, R), (B, L), (B, R)\}.$$

Under a Markov policy,

$$P(A_{n+1} = a' \mid S_{n+1} = s', \text{history}) = \pi_{n+1}(a' \mid s').$$

Therefore,

$$\begin{aligned} P^\pi((S_{n+1}, A_{n+1}) = (s', a') \mid (S_n, A_n) = (s, a)) \\ = P(s' \mid s, a)\pi_{n+1}(a' \mid s'). \end{aligned}$$

This is the transition probability of the state-action process.

Order the state-action pairs as $(A, L), (A, R), (B, L), (B, R)$. For example,

$$P((A, L) \mid (A, L)) = P(A \mid A, L)\pi(L \mid A) = 0.8(0.7) = 0.56.$$

Similarly,

$$P((A, R) \mid (A, L)) = 0.8(0.3) = 0.24,$$

$$P((B, L) \mid (A, L)) = 0.2(0.4) = 0.08,$$

$$P((B, R) \mid (A, L)) = 0.2(0.6) = 0.12.$$

Hence the complete matrix is

$$P_{SA}^\pi = \begin{pmatrix} 0.56 & 0.24 & 0.08 & 0.12 \\ 0.21 & 0.09 & 0.28 & 0.42 \\ 0.42 & 0.18 & 0.16 & 0.24 \\ 0.07 & 0.03 & 0.36 & 0.54 \end{pmatrix}.$$

5. Is the process $\{(S_n, A_n)\}$ time homogeneous?

The policy is stationary:

$$\pi_n(a|s) = \pi(a|s) \quad \forall n.$$

The environment is also time homogeneous:

$$P(S_{n+1} = s' \mid S_n = s, A_n = a) = P(s' \mid s, a).$$

Therefore,

$$P((S_{n+1}, A_{n+1}) = (s', a') \mid (S_n, A_n) = (s, a))$$

does not depend on n .

Indeed,

$$P((s', a') \mid (s, a)) = P(s' \mid s, a)\pi(a' \mid s').$$

Thus,

$$\{(S_n, A_n)\}_{n \geq 0} \text{ is a time-homogeneous Markov chain.}$$

Remember:

time-homogeneous environment + stationary policy
 $\Rightarrow$ time-homogeneous state-action process.

2.2 (Non-Stationary Markov Policy) Consider

$$\pi_{2k}(L | s) = 1, \quad \pi_{2k+1}(R | s) = 1.$$

Thus the policy alternates: $L, R, L, R, \dots$ irrespective of the state.

The action at time n is determined by n , so it is still a **Markov policy**, because it does not depend on the past history. However, it is **non-stationary**, because

$$\pi_0 \neq \pi_1.$$

Very important distinction:

$$\text{Markov} \neq \text{stationary}.$$

A Markov policy can depend on time:

$$\pi_n(a|s).$$

A stationary policy additionally satisfies

$$\pi_n(a|s) = \pi(a|s) \quad \forall n.$$

1. Let $Y_n = (S_n, A_n)$. We need to show

$$P(Y_{n+1} | Y_n, Y_{n-1}, \dots, Y_0) = P(Y_{n+1} | Y_n).$$

Since the environment is Markov,

$$P(S_{n+1} = s' | S_n = s, A_n = a, \text{past}) = P(s' | s, a).$$

Also, under the deterministic Markov policy,

$$P(A_{n+1} = a' | S_{n+1} = s', \text{past}) = \pi_{n+1}(a' | s').$$

Therefore,

$$P(Y_{n+1} = (s', a') | Y_n = (s, a), \text{past}) = P(s' | s, a) \pi_{n+1}(a' | s').$$

The right-hand side depends only on $Y_n = (s, a)$ and time n .

Hence,

$$\{(S_n, A_n)\}_{n \geq 0} \text{ is a time-inhomogeneous Markov chain.}$$

2. **Transition Matrix for the State–Action Process** Consider the state–action space

$$\mathcal{X} = \{(A, L), (A, R), (B, L), (B, R)\}.$$

We use the ordering

$$(A, L), \quad (A, R), \quad (B, L), \quad (B, R).$$

The transition probability for the state–action process is

$$P_n((s', a') \mid (s, a)) = P(s' \mid s, a) \pi_{n+1}(a' \mid s').$$

At time $n = 0$ Since $n = 0$ is even, the policy is

$$\pi_0(L \mid s) = 1.$$

Therefore, at time $n = 0$, only the state–action pairs

$$(A, L), \quad (B, L)$$

can occur. The pairs (A, R) and (B, R) have probability zero, so the corresponding rows of the conditional transition matrix are undefined.

At time $n = 1$, the policy chooses R with probability one:

$$\pi_1(R \mid s) = 1.$$

For (A, L) ,

$$P((A, R) \mid (A, L)) = P(A \mid A, L) \pi_1(R \mid A) = 0.8,$$

$$P((B, R) \mid (A, L)) = P(B \mid A, L) \pi_1(R \mid B) = 0.2.$$

Thus, the row corresponding to (A, L) is

$$(0, 0.8, 0, 0.2).$$

Similarly, for (B, L) ,

$$P((A, R) \mid (B, L)) = P(A \mid B, L) \pi_1(R \mid A) = 0.6,$$

$$P((B, R) \mid (B, L)) = P(B \mid B, L) \pi_1(R \mid B) = 0.4.$$

Thus, the row corresponding to (B, L) is

$$(0, 0.6, 0, 0.4).$$

Therefore, the transition matrix at time $n = 0$ is

$$P_0 = \begin{pmatrix} 0 & 0.8 & 0 & 0.2 \\ \text{undefined} & \text{undefined} & \text{undefined} & \text{undefined} \\ 0 & 0.6 & 0 & 0.4 \\ \text{undefined} & \text{undefined} & \text{undefined} & \text{undefined} \end{pmatrix}$$

where the rows and columns are ordered as

$$(A, L), \quad (A, R), \quad (B, L), \quad (B, R).$$

At time $n = 1$ Since $n = 1$ is odd, the policy is

$$\pi_1(R | s) = 1.$$

Therefore, at time $n = 1$, only the state–action pairs

$$(A, R), \quad (B, R)$$

can occur. The pairs (A, L) and (B, L) have probability zero, so the corresponding rows of the conditional transition matrix are undefined.

At time $n = 2$, the policy chooses L with probability one:

$$\pi_2(L | s) = 1.$$

For (A, R) ,

$$P((A, L) | (A, R)) = P(A | A, R)\pi_2(L | A) = 0.3,$$

$$P((B, L) | (A, R)) = P(B | A, R)\pi_2(L | B) = 0.7.$$

Thus, the row corresponding to (A, R) is

$$(0.3, 0, 0.7, 0).$$

Similarly, for (B, R) ,

$$P((A, L) | (B, R)) = P(A | B, R)\pi_2(L | A) = 0.1,$$

$$P((B, L) | (B, R)) = P(B | B, R)\pi_2(L | B) = 0.9.$$

Thus, the row corresponding to (B, R) is

$$(0.1, 0, 0.9, 0).$$

Therefore, the transition matrix at time $n = 1$ is

$$P_1 = \begin{pmatrix} \text{undefined} & \text{undefined} & \text{undefined} & \text{undefined} \\ 0.3 & 0 & 0.7 & 0 \\ \text{undefined} & \text{undefined} & \text{undefined} & \text{undefined} \\ 0.1 & 0 & 0.9 & 0 \end{pmatrix}$$

where the rows and columns are ordered as

$$(A, L), \quad (A, R), \quad (B, L), \quad (B, R).$$

Thus, the defined rows alternate with time:

Time	Possible state–action pairs
$n = 0$	$(A, L), (B, L)$
$n = 1$	$(A, R), (B, R)$
$n = 2$	$(A, L), (B, L)$
$\vdots$	$\vdots$

3. Homogeneous or Inhomogeneous?

At $n = 0$,

$$K_0((A, L), (A, R)) = 0.8.$$

At $n = 1$,

$$K_1((A, L), (A, L)) = 0.8.$$

Thus,

$$K_0 \neq K_1.$$

Therefore, the transition probabilities depend on time.

Hence,

$$\{(S_n, A_n)\}_{n \geq 0} \text{ is time-inhomogeneous.}$$

The reason is not that the environment changes. The environment remains time homogeneous.

The source of inhomogeneity is the policy:

$$\pi_0 \neq \pi_1.$$

Remember:

A time-homogeneous environment does *not* automatically imply a time-homogeneous controlled process.
The policy can introduce time dependence.

2.3 (History-Dependent Policy) Now consider

$$A_n = \begin{cases} L, & S_{n-1} = A, \\ R, & S_{n-1} = B, \end{cases} \quad n \geq 1.$$

The action at time n depends on the *previous state*.

Therefore, $A_n \neq \pi_n(S_n)$ in general.

Instead,

$$A_n = f(S_{n-1}).$$

This is a history-dependent policy because the previous state is part of the history.

Key distinction:

Policy type	Action depends on
Markov	S_n
History-dependent	$S_0, A_0, \dots, A_{n-1}, S_n$

The current state alone does not determine the action in this example.

1. We want

$$P(S_{n+1} = B \mid S_n = A, S_{n-1} = A).$$

Since $S_{n-1} = A$, the policy chooses $A_n = L$. Therefore,

$$\begin{aligned} &P(S_{n+1} = B \mid S_n = A, S_{n-1} = A) \\ &= P(S_{n+1} = B \mid S_n = A, A_n = L) \\ &= P(B \mid A, L) \\ &= \boxed{0.2}. \end{aligned}$$

2. Now consider

$$P(S_{n+1} = B \mid S_n = A, S_{n-1} = B).$$

Since $S_{n-1} = B$, the policy chooses $A_n = R$. Therefore,

$$\begin{aligned} &P(S_{n+1} = B \mid S_n = A, S_{n-1} = B) \\ &= P(S_{n+1} = B \mid S_n = A, A_n = R) \\ &= P(B \mid A, R) \\ &= \boxed{0.7}. \end{aligned}$$

3. For the state process to be Markov, we would need

$$P(S_{n+1} = B \mid S_n = A, S_{n-1}) = P(S_{n+1} = B \mid S_n = A).$$

But we obtained

$$P(S_{n+1} = B \mid S_n = A, S_{n-1} = A) = 0.2$$

while

$$P(S_{n+1} = B \mid S_n = A, S_{n-1} = B) = 0.7.$$

Thus, knowing the previous state changes the distribution of the next state even after $S_n = A$ is known.

Therefore,

$$\boxed{\{S_n\}_{n \geq 0} \text{ is not a Markov chain.}}$$

Best way to disprove Markov property:

Find two histories with the same current state but different conditional next-state probabilities.

4. Interestingly, the state-action process is Markov.

Recall

$$A_n = \begin{cases} L, & S_{n-1} = A, \\ R, & S_{n-1} = B. \end{cases}$$

Therefore, observing A_n tells us the previous state:

$$A_n = L \Rightarrow S_{n-1} = A,$$

and

$$A_n = R \Rightarrow S_{n-1} = B.$$

Hence, from the current pair (S_n, A_n) , the relevant previous state is already encoded in A_n . Moreover,

$$A_{n+1} = \begin{cases} L, & S_n = A, \\ R, & S_n = B. \end{cases}$$

Thus, A_{n+1} depends only on the current state S_n . Therefore,

$$\boxed{\{(S_n, A_n)\}_{n \geq 0} \text{ is a Markov chain.}}$$

5. Which assumption of Theorem 1 is violated in this example? Explain why this assumption is essential. The theorem assumes that the policy is **Markov**:

$$\boxed{\pi_n(a \mid \text{history}) = \pi_n(a \mid S_n).}$$

In the present example,

$$A_n = \begin{cases} L, & S_{n-1} = A, \\ R, & S_{n-1} = B. \end{cases}$$

Thus,

$$A_n$$

depends on S_{n-1} , not only on S_n .

Therefore the Markov-policy assumption is violated.

$$\boxed{\text{The policy is history-dependent, not Markov.}}$$

This is exactly why the state process $\{S_n\}$ fails to be Markov.